

A decision matrix below translates recovery-model choices into an operational playbook you can apply across SQL Server estates. It reflects workload category, RPO/RTO targets, logging behavior, backup cadence, and operational risks.

SQL Server Recovery Model Decision Matrix

1. Summary Matrix

Workload Type	Business Criticality	RPO Requirement	Preferred Recovery Model	Restore Capability	Logging Behavior	Operational Notes
High-volume OLTP (Banking, ERP, eCommerce)	Very high	Near-zero data loss	Full	Point-in-time restore	Fully logged	Requires frequent log backups; monitor log growth
Mixed workload with periodic heavy bulk loads (ETL, DW staging, nightly loads)	High	Low-to-mid	Bulk-Logged (temporarily)	No PITR during bulk window	Minimal logging for bulk ops	Switch back to Full; take log backups before and after bulk load
Reporting/Analytics (Read-only, infrequent changes)	Low	Can tolerate some data loss	Simple	Restore to last full/diff only	Log auto-truncated	Lower admin overhead
Dev/Test/QA Environments	Very low	Not relevant	Simple	Limited restore	Log auto-truncated	Easy reset using restores or snapshots
Data warehouses (large loads + scheduled ETL)	Medium	Moderate	Bulk-Logged (during ETL) + Full (rest of time)	Mixed; PITR limited	Minimal logging during ETL	Good for managing log growth
Archival or Static Data	Very low	Can lose data since static	Simple	Only full/diff	Minimal logging	Backups spaced out; log remain small

2. Detailed Decision Criteria

A. Critical Workloads (Financial, Healthcare, Inventory)

- RPO target: < 1 minute.
- RTO target: Aggressive.
- Required: Zero or near-zero data loss.
- Recovery Model: **Full**.
- Why: Supports point-in-time restore, log shipping, AlwaysOn AG synchronous mode.
- Backup Strategy:
 - Full: daily or weekly
 - Differential: daily or frequent
 - Log backups: every 5–15 minutes
- Risks:
 - Large log file growth if log backups not scheduled
 - Many VLFs slow recovery

- Mitigation:
 - Use frequent log backups and VLF monitoring
 - TempDB optimization for heavy workloads

B. Bulk Data Processing or ETL Pipelines

- RPO target: A few minutes or hours.
- RTO target: Moderate.
- Data behavior: Heavy loads at scheduled windows.
- Recovery Model: **Bulk-Logged (only during bulk operations)**.
- Why: Reduces log generation for operations like:
 - Bulk Insert
 - Index rebuild
 - Select Into
 - Large ETL batches
- Backup Strategy:
 - Log backup immediately before bulk load
 - Log backup immediately after bulk load
 - Switch back to Full after ETL completes
- Risks:
 - No point-in-time restore inside bulk-logged window
- Mitigation:
 - Strict scheduling and documentation
 - Automated pre- and post-load backup job

C. Standard Reporting or BI Systems

- RPO target: Hours.
- RTO target: Flexible.
- Writes: Minimal.
- Recovery Model: **Simple** or **Full** (if analytics on OLTP replica).
- Why:
 - Simple reduces log management complexity
 - Adequate for non-critical periodic loads
- Backup Strategy:
 - Full backup daily/weekly
 - Differential daily
- Risks:
 - All transactions since last backup are lost if failure occurs
- Mitigation:
 - Backup frequency matched to data refresh cycles

D. Development, Testing, Training Environments

- RPO target: Not required.
- Writes: Frequent but non-critical.
- Recovery Model: **Simple**.
- Why:
 - Eliminates log management overhead

- Easy to reset with restore from production snapshot
- Backup Strategy:
 - Full backup weekly or as automation requires
- Risks:
 - None significant
- Mitigation:
 - Refresh periodically to avoid large data drift

E. Large Data Warehouses (TB-scale fact tables)

- RPO target: Medium.
- Workload: Heavy nightly ETL + large fact table operations.
- Recovery Model:
 - **Bulk-Logged during ETL**
 - **Full** during normal operation
- Why:
 - Minimally logged operations drastically reduce log size
 - PITR still available outside the bulk window
- Backup Strategy:
 - Pre-ETL log backup
 - Post-ETL log backup
 - Weekly full, daily differential
- Risks:
 - PITR blocked during bulk
- Mitigation:
 - Document ETL windows
 - Automate switching model if used

3. Recovery Model Selection Flowchart (Text Version)

1. Does the business require recovery to an exact point in time?
 - Yes → Use **Full**.
 - No → Go to next question.
2. Does the workload include large bulk operations that cause huge log growth?
 - Yes → Use **Bulk-Logged** (only during operations), otherwise Full.
 - No → Go to next question.
3. Is the database non-critical, dev/test, or acceptable to lose data since last full/diff backup?
 - Yes → Use **Simple**.
 - No → Use **Full**.

4. Backup and Restore Strategy Alignment

Recovery Model	Backup Chain Requirements	Supported Restore Types	Not Supported
Full	Full + Diff + Log	Point-in-time restore, marked transactions	None
Bulk-Logged	Full + Diff + Log	Restore to end of last log backup	PITR during bulk ops
Simple	Full + Diff	Restore to last backup only	Log backups, PITR

5. RPO/RTO Alignment Quick Reference

RPO Target	Recommended Recovery Model
Seconds	Full
< 15 minutes	Full
15 minutes to 4 hours	Full or Bulk-Logged
> 4 hours (non-critical)	Simple
No RPO requirement	Simple

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Below is a structured DBA runbook comparing **Full**, **Bulk-Logged**, and **Simple** recovery models. It is formatted as a practical operational document that a production DBA team can use for change management, DR planning, and incident response.

SQL Server Recovery Models

DBA Runbook and Operational Comparison Guide

This runbook defines how each recovery model behaves, when to use it, how to manage backups, and how to safely switch models in production environments.

1. Purpose of this Runbook

- Establish a standardized operational understanding of SQL Server recovery models.
- Provide decision guidance for capacity planning, backup strategy, and DR requirements.
- Document safe operational procedures for switching recovery models.
- Reduce risk of broken log chains, uncontrolled log growth, and data-loss exposure.

2. Recovery Model Overview

2.1 Full Recovery Model

Definition

SQL Server logs all transactions in full detail and does not truncate the log until a log backup occurs.

Key Characteristics

- Supports point-in-time restore.
- Requires transaction log backups to maintain log chain.
- Log file grows indefinitely without regular log backups.
- Supports Log Shipping, Availability Groups, Mirroring.

When to Use

- OLTP systems
- Critical databases with strict RPO/RTO
- Systems requiring PITR

2.2 Bulk-Logged Recovery Model

Definition

A variant of Full recovery that allows minimal logging for certain bulk operations.

Key Characteristics

- Reduces logging on large operations.
- Log backups still required.
- Point-in-time restore is **not** possible during bulk-logged operations.
- Used temporarily for data-load windows.

When to Use

- Heavy index maintenance
- Bulk imports (BCP, Bulk Insert, SSIS)
- DW ETL pipelines

2.3 Simple Recovery Model

Definition

SQL Server automatically truncates the transaction log after each checkpoint.

Key Characteristics

- No log backups possible.
- Only restore to last full or differential backup.
- Automatically controls log growth.
- Minimal administration overhead.

When to Use

- Dev/Test environments
- Reporting DBs where data refresh is periodic
- Non-critical databases

3. Comparison Matrix

Feature / Capability	Full	Bulk-Logged	Simple
Point-in-time restore	Yes	No (during bulk ops)	No
Transaction log backups	Required	Required	Not allowed
Log growth control	Manual (via T-log backups)	Manual	Automatic
Bulk operation logging	Fully logged	Minimally logged	Fully logged
Supports Log Shipping	Yes	Yes	No
Supports AlwaysOn AG	Yes	Yes	No
Supports Database Mirroring	Yes	Yes	No
Best for OLTP systems	Yes	No	Rarely
Best for ETL/Data load	Sometimes	Yes	No
Best for Dev/Test	No	No	Yes

4. Operational Procedures**4.1 Validate Current Recovery Model**

```
SELECT name, recovery_model_desc
FROM sys.databases;
```

4.2 Switching to Full Recovery Model (Safe Procedure)**Step 1: Switch model**

```
ALTER DATABASE YourDB SET RECOVERY FULL;
```

Step 2: Immediately take a full backup to initialize the log chain

```
BACKUP DATABASE YourDB TO DISK = 'D:\Backups\YourDB_Full.bak';
```

Step 3: Implement log backup schedule

Minimum recommended frequency: every 15 minutes.

4.3 Switching to Bulk-Logged Model (ETL Window)**Step 1: Pre-ETL log backup**

```
BACKUP LOG YourDB TO DISK = 'D:\Backups\YourDB_PreETL.trn';
```

Step 2: Switch model

```
ALTER DATABASE YourDB SET RECOVERY BULK_LOGGED;
```

Step 3: Run bulk operations

(Minimal logging applied to eligible operations.)

Step 4: Post-ETL log backup

```
BACKUP LOG YourDB TO DISK = 'D:\Backups\YourDB_PostETL.trn';
```

Step 5: Revert to Full

```
ALTER DATABASE YourDB SET RECOVERY FULL;
```

4.4 Switching to Simple Recovery Model**Step 1: Assess production impact**

- PITR will permanently be unavailable.
- Log shipping / AG / Mirroring will stop functioning.

Step 2: Switch model

```
ALTER DATABASE YourDB SET RECOVERY SIMPLE;
```

Step 3: Shrink log only if reclaiming space is necessary

```
DBCC SHRINKFILE (YourDB_log, TARGET_SIZE_MB);
```

Step 4: Implement full + differential backups

Frequencies normally once daily (full) and periodic differential backups.

5. Backup Strategy by Recovery Model**5.1 Full Recovery Model**

- Full backup: Weekly
- Differential backup: Daily
- Log backups: Every 5–15 minutes
- Additional considerations:
 - Monitor VLF count
 - Maintain log file size appropriately

5.2 Bulk-Logged Recovery Model

- Same schedule as Full
- Mandatory: Log backup immediately after every bulk operation
- Note: PITR not available during bulk ops

5.3 Simple Recovery Model

- Full backup: Daily or weekly
- Differential backup: Daily (optional)
- No log backups
- Log size stable due to automatic truncation

6. Disaster Recovery Implications

DR Requirement	Full	Bulk-Logged	Simple
Restore to exact timestamp	Yes	No	No
Restore after major data corruption	Yes	Partial	Partial
Restore with minimal data loss	Yes	Sometimes	No
DR replication supported	Yes	Yes	No

7. Monitoring and Alerting Requirements

7.1 Full/Bulk-Logged

- Log file growth > threshold
- Log backups not running
- VLF count > 10,000
- Replication/AG log send queue growth
- Disk free space for logs

7.2 Simple

- Less monitoring required
- Monitor normal disk utilization

8. Risk Register

Risk	Model Impacted	Mitigation
Log file grows to fill disk	Full, Bulk-Logged	Schedule frequent log backups
Broken log chain	Full, Bulk-Logged	Always take backup after switching models
No PITR	Bulk-Logged, Simple	Ensure business accepts RPO limitations
ETL creates massive logs	Full	Use Bulk-Logged for ETL windows
DR failovers impossible	Simple	Avoid Simple for production

9. Governance and Change Management

- Any switch from Full to Simple must be audited and approved.
- Document the start/end time of Bulk-Logged windows.
- Capture pre- and post-change backups.
- Maintain version-controlled notes for every recovery model change.

10. Summary Recommendations

- Use **Full** for all critical production OLTP systems.
- Use **Bulk-Logged** only during large planned bulk operations.
- Use **Simple** for non-critical, dev/test, read-only, or archival databases.
- Align every recovery model decision with RPO, RTO, and DR strategy.
- Test restore chains quarterly using actual backups.

One-page comparison sheet for SQL Server recovery models. It is concise, decision-oriented, and suitable for architects, managers, and senior DBAs.

SQL Server Recovery Models: One-Page Executive Summary

Purpose

Define how SQL Server logs transactions, what restore options are available, and how each model aligns with business RPO/RTO and operational needs.

1. Recovery Models at a Glance

Model	Logging Behavior	Restore Capability	Backup Requirements	Ideal Use Cases
Full	Fully logs all transactions; log truncation only after log backup	Point-in-time restore	Full + Differential + Log backups	Critical OLTP, financial systems, zero data loss environments
Bulk-Logged	Minimally logs eligible bulk operations	No point-in-time restore during bulk ops	Full + Differential + Log backups	Data loads, index rebuilds, ETL windows, DW staging
Simple	Log auto-truncated at checkpoint	Restore only to last full or differential backup	Full + Differential backups	Dev/Test, reporting, non-critical, read-mostly databases

2. Key Capabilities Comparison

Capability	Full	Bulk-Logged	Simple
Point-in-time restore	Yes	No (during minimal logging)	No
Log backups	Required	Required	Not allowed
Log growth control	Manual	Manual	Automatic
Supports HA/DR (Log Shipping, AG, Mirroring)	Yes	Yes	No
Minimal logging for bulk operations	No	Yes	No

3. When to Choose Each Model

Full

Select when data loss is unacceptable and precise recovery is required.

- Frequent transactional workloads
- High-availability architectures
- Regulatory or audit-driven requirements

Bulk-Logged

Use temporarily for scheduled bulk operations to reduce log volume.

- Large index maintenance
- Bulk Import / BCP / SELECT INTO
- Heavy ETL pipelines

Simple

Choose when log backups are unnecessary or overhead is not justified.

- Development and test systems
- Reporting/analytics refreshed periodically
- Non-critical environments

4. Backup Strategy Alignment

Model	Recommended Cadence
Full	Full weekly, differential daily, log backups every 5–15 minutes
Bulk-Logged	Same as Full, plus log backup immediately before and after bulk ops
Simple	Full daily or weekly, differential daily (optional)

5. Risks and Mitigations

Risk	Model	Mitigation
Log grows to consume disk	Full/Bulk-Logged	Schedule frequent log backups; size log correctly
Broken log chain	Full/Bulk-Logged	Always take backup immediately after model switch
No PITR	Bulk-Logged/Simple	Validate business RPO; avoid Simple for production
ETL causing huge log growth	Full	Temporarily switch to Bulk-Logged

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